

Question Paper

Exam Date & Time: 03-May-2025 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FOURTH SEMESTER B.TECH. (INFORMATION TECHNOLOGY) DEGREE EXAMINATIONS - MAY 2025
SUBJECT: ICT 2223 / IT_2223 - EMBEDDED SYSTEMS

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) Explain the fundamental concepts of computer architecture with reference to RISC-based processors. (5)
(a) Compare the RISC and CISC instructions set architectures, and explain the key differences in terms of complexity and performance.
(b) Explain different types of addressing modes used in ARM architecture and their significance in program execution.
- 1B) Write an ARM Assembly Language Program using recursion to calculate the sum of squares of the first 'n' natural numbers. (3)
- 1C) Explain the role of the Nested Vectored Interrupt Controller (NVIC) in the LPC176x microcontroller. How does NVIC handle vectored interrupts, and how are interrupt vectors stored and accessed in the Interrupt Vector Table (IVT)? (2)
- 2A) Develop an embedded C program to turn ON the LED connected to P0.4 whenever the input at P2.10 (EINT0, function 1) is HIGH and turn OFF the LED if the input at P2.10 is LOW. (5)
- 2B) Develop an ALP to find GCD of two 32-bit number. (3)
- 2C) Write a Embedded C program to rotate the stepper motor in the clockwise direction. (2)
- 3A) Assume columns of a 2x2 matrix keyboard are connected to P2.10 and P2.11 and rows are connected to P2.12 and P2.13. Write an embedded C program to display the keycode of key pressed on LEDs connected at P2.14 and P2.15 using GPIO interrupts (5)
- 3B) Contrast the ways that ADC and timer interrupts clearing. Demonstrate with embedded C statements to clear the interrupt generated by the ADC channel ADC0.6 (3)
- 3C) Demonstrate the followings (2)
i) Instruction RRXS
ii) Data format DCB
- 4A) Write an Embedded C program for Keil using an LP C1768 microcontroller that uses a hardware timer (TIM2) configured for PWM mode to vary the intensity of an LED. (5)
- 4B) Using DAC, generate a variable frequency positive raising sawtooth waveform. Use ROW-0 of keyboard for frequency variation. (3)
- 4C) Distinguish between Single edge and double edge PWM. (2)
- 5A) Write an embedded C program Turn ON the LED connected to P1.23 whenever the switch connected to P2.12 (EINT2, Function-01) is pressed LED remains ON as long as the switch is pressed. (Assume, by default logic-0 is INPUT and when the switch is pressed, Logic-1 is INPUT). (5)
- 5B) Write an Embedded C program to read an analog input voltage from AD0.5 (P1.31, SF3) and convert it to a digital value using the on-chip ADC. Based on the converted digital value, control LEDs connected to P0.4 and P0.5 as follows: (3)
i) If the digital value is less than 1024, turn ON the LED at P0.4.

- ii) If the digital value is greater than 3072, turn ON the LED at P0.5.
 iii) If the digital value is between 1024 and 3072 (inclusive), keep both LEDs OFF.

5C) Explain the role of the MAX232 IC in serial communication between a microcontroller and an RS-232 port (2)

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Event Name		Subject	
MIT MPL and BLR- BTech-M Sc-MCA - II-IV and VI Sem - End Semester Examination - Apr-May 2025		IT-2223:Embedded Systems	
#	Course Outcome	Description	Marks

No records found / CO Level Description not defined

* **NA** - CO Not Selected

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